From Correlation to Mechanism: Unifying Causal Inference and Deep Learning through Identifiability and Indefinite Data Paradigms

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The integration of causal inference with deep learning marks a paradigmatic shift from extracting statistical correlations to modeling underlying generative mechanisms. This synthesis addresses the fragility of standard neural networks under distribution shifts by embedding structural constraints directly into flexible representation spaces. Central to this evolution is the transition from restrictive linear assumptions to nonparametric identifiability, where environmental heterogeneity and soft interventions enable the recovery of latent causal variables without strong parametric priors [96]. By leveraging the Principle of Independent Causal Mechanisms, modern frameworks achieve robust domain generalization, distinguishing invariant causal relations from spurious associations [289]. Furthermore, the field is expanding beyond definite tabular data to encompass "indefinite" paradigms-such as dialogue and video-where multi-structure representations challenge traditional causal discovery [6]. This document synthesizes theoretical advances in identifiability, architectural innovations like differentiable DAG learning, and emerging applications in biomedical and temporal systems. It argues that the future of artificial intelligence relies on unifying expressivity with causal rigor, transforming deep learning from a predictive tool into a robust engine for counterfactual reasoning and scientific discovery.

Introduction: The Paradigm Shift from Correlation to Causal Mechanism

The integration of causal inference with deep learning represents a fundamental evolution in machine learning, addressing the intrinsic limitations of standard neural networks that excel at identifying statistical correlations but frequently fail to generalize under distribution shifts, lack interpretability, and cannot perform counterfactual reasoning. While deep learning has achieved remarkable success in extracting patterns from high-dimensional data, these models often rely on spurious correlations that do not hold when the data distribution changes, a vulnerability attributed to confounding effects and the absence of underlying mechanistic understanding [1]. Consequently, the emerging field of Causal Deep Learning (CDL) seeks to structure and encode causal knowledge within the flexible representation space of deep learning models to achieve more informed, robust, and general predictions [9]. This paradigm shift moves beyond mere association to uncover the generative mechanisms that drive data, enabling systems that are not only predictive but also capable of answering "what-if" questions and supporting interventions [41].

The theoretical foundation of this shift rests on the distinction between statistical associations and causal relationships, often formalized through frameworks such as Pearl's Structural Causal Model (SCM), the Rubin Causal Model, and Granger Causality [1]. Traditional causal learning methods have long struggled with challenges such as high-dimensional unstructured variables, combinatorial optimization problems, and unobserved confounders [2]. Deep learning offers innovative solutions to these persistent issues by providing strong representational capabilities for unstructured data like images and text, powerful fitting abilities to approximate complex non-linear functions, and the capacity to explicitly or implicitly model data generation mechanisms [2]. For instance, deep generative models can be employed to learn structural causal models that answer counterfactual queries using observational data, a task that was previously intractable for many parametric methods [64]. By utilizing normalizing flows and variational inference, researchers have developed frameworks capable of tractable inference of exogenous noise variables, a crucial step for counterfactual reasoning that validates the potential of deep SCMs to operate across all three levels of Pearl's ladder of causation: association, intervention, and counterfactuals [139].

A critical aspect of this synthesis is the move towards causal representation learning, which aims to discover high-level causal variables from low-level observations, thereby addressing the "black box" nature of standard deep neural networks [10]. This approach posits that by learning representations that align with the true causal factors of variation, models can achieve better disentanglement, interpretability, and transferability [3]. The principle of independent causal mechanisms (ICM) further supports this direction, suggesting that causal mechanisms remain invariant across different environments, whereas statistical correlations may shift [289]. Leveraging this principle, deep learning frameworks can be designed to focus on relations that remain stable across environments, effectively ignoring unstable spurious correlations and improving out-of-distribution generalization [289]. Recent theoretical advances have established sufficient and necessary conditions for the identifiability of latent causal representations under distribution shifts, demonstrating that leveraging multi-environment data allows for the recovery of true causal structures even in non-linear additive noise models [96]. Furthermore, frameworks like ICM-VAE have shown that enforcing causal disentanglement priors leads to representations that are robust to interventions and compatible with counterfactual generation [11].

The scope of CDL has expanded beyond traditional tabular data to encompass complex, non-statistical data types, necessitating a redefinition of causal data paradigms. Researchers now categorize data into definite, semi-definite, and indefinite paradigms, where indefinite data-such as dialogues and videos-presents unique challenges due to conflicts between causal structures and representations [6]. To address these complexities, new baselines and probabilistic frameworks are being developed to handle the multi-structure and multi-value nature of indefinite data, incorporating causal strength as a latent variable to overcome previous infeasibilities [168]. In the temporal domain, the field has moved from static graph discovery to dynamic causal discovery, with methods like UnCLe and CausalFormer utilizing deep learning to capture evolving temporal causality and non-linear dependencies in time-series data [245; 136]. These developments underscore the transition from passive pattern recognition to active causal understanding, positioning causal deep learning as a cornerstone for building robust, fair, and interpretable artificial intelligence systems capable of navigating the complexities of the real world [41]. Emerging architectures such as Causal-Symbolic Meta-Learning (CSML) and Hierarchical Causal Primitive Dynamic Composition Networks (HCP-DCNet) further illustrate this trajectory by inducing causal world models that support few-shot generalization and compositional reasoning, bridging the gap between continuous physical dynamics and discrete symbolic inference [280; 87].

The Limitations of Associational Learning and the I.I.D. Assumption

Standard deep learning paradigms operate fundamentally on the extraction of statistical correlations from high-dimensional data, an approach that has yielded remarkable success in tasks where training and testing distributions align. However, this reliance on associational learning renders models inherently fragile when deployed in real-world scenarios where the independent and identically distributed (i.i.d.) assumption is violated. As noted in [1], deep learning models struggle to generalize outside their initial distribution because they often latch onto spurious correlations driven by confounding effects rather than identifying the underlying causal mechanisms. This limitation is not merely a matter of insufficient data or model capacity; it is a structural deficiency of optimizing for predictive accuracy without regard for the data-generating process. When the environment shifts, these unstable correlations break down, leading to catastrophic performance degradation. The core issue lies in the inability of standard supervised learning to distinguish between invariant causal mechanisms and environment-specific noise, a failure that persists even as model scale increases.

The theoretical disconnect between prediction and causation is starkly illustrated by the "predictive-causal gap." [91] presents an impossibility theorem showing that when environment modes are slower or less noisy than system modes, the optimal encoder for a predictive objective will track the environment rather than the system. This results in a representation that is predictively optimal but causally blind, a failure mode that scaling model capacity cannot close. This finding challenges the prevailing assumption in self-supervised learning that minimizing prediction error inherently recovers causal structure. In contrast, the principle of independent causal mechanisms (ICM) offers a pathway to robustness by leveraging distribution shifts across different environments during training. [289] argues that by guiding models to focus on relations that remain invariant across environments, one can effectively ignore unstable associations. Theoretical and experimental evidence suggests that neural networks trained with objectives derived from the ICM principle recover stable relations that correspond to true causal mechanisms, thereby generalizing well to unseen scenarios where traditionally trained models fail. This approach is further supported by [51], which extends invariance guarantees to nonlinear settings by assuming a flexible conditionally non-factorized prior, enabling the discovery of direct causes of the target even in complex, high-dimensional spaces.

Furthermore, the challenge of identifiability complicates the transition from correlation to causation, particularly in latent variable settings. Without specific inductive biases or access to interventional data, disentangling latent causal factors from observational data is often impossible. [96] establishes that determining the types of distribution shifts contributing to identifiability is critical, providing sufficient and necessary conditions for identifying latent additive noise models by leveraging seen distribution shifts. This extends previous findings limited to linear or polynomial models, addressing the numerical instability inherent in high-degree polynomial approximations. Similarly, [130] highlights that standard methods relying on statistical independence fail to capture causal dependencies without strong inductive biases, such as the Additive Noise Model (ANM), which restricts admissible transformations and resolves component-wise indeterminacy. These works collectively underscore that without explicit constraints or multi-environment data, deep learning models cannot uniquely identify the causal structure necessary for robust generalization.

The consequences of ignoring causal structure extend beyond simple accuracy drops; they manifest as a lack of reliability in counterfactual reasoning and intervention, particularly in complex data structures like graphs and time series. [75] clarifies that many ad hoc notions of "invariance" in domain adaptation contradict each other because they fail to account for the true underlying causal structure. Robustness to domain shifts is only achievable when the learned invariant representations correspond to the actual causal mechanisms compatible with the real-world domain shift. In the realm of graph neural networks, [192] addresses how traditional aggregation methods amplify spurious patterns, limiting robustness under distribution shifts. Their approach performs causal interventions on graph structures to identify and preserve causally relevant connections, thereby learning invariant node representations. Additionally, the challenge of temporal dynamics is addressed in [67], which notes that existing evolving domain generalization methods often suffer from spurious correlations by modeling only dependencies across domains without accounting for causal mechanism drifts. They propose a time-aware structural causal model to capture dynamic causal factors, contrasting with methods like [245] that utilize uncoupler and recoupler networks to disentangle input time series and learn inter-variable dependencies via auto-regressive dependency matrices.

Ultimately, the literature converges on the conclusion that the i.i.d. assumption is a significant bottleneck for the deployment of deep learning in dynamic, open-world settings. [61] emphasizes that while deep causal models map covariates to representation spaces to estimate counterfactuals unbiasedly, the field must move beyond simple associative mapping. The integration of causal inductive biases, whether through the principle of independent causal mechanisms, invariant risk minimization, or explicit structural modeling, is not optional but essential for creating systems that possess the robustness and adaptability required for genuine intelligence. As [10] succinctly puts it, the discovery of high-level causal variables from low-level observations is a central problem that, if unsolved, will continue to limit the generalization capabilities of artificial intelligence. The shift from purely predictive objectives to those grounded in causal discovery represents a fundamental reorientation of machine learning research, necessary to overcome the structural limitations of associational learning.

Theoretical Frameworks Unifying Causality and Deep Learning

The synthesis of causal inference and deep learning represents a paradigmatic shift from purely correlational pattern recognition to the modeling of underlying generative mechanisms. This unification is primarily grounded in the extension of Structural Causal Models (SCMs) and the Potential Outcome Framework into flexible neural architectures. While traditional SCMs provide a rigorous graphical formalism for representing cause-effect relationships through directed acyclic graphs and structural equations, they often struggle with high-dimensional, unstructured data and complex non-linear functional forms. Conversely, deep neural networks offer universal approximation capabilities but lack inherent causal semantics, rendering them susceptible to spurious correlations and distribution shifts. Recent theoretical advancements have sought to bridge this gap by defining Neural Causal Models (NCMs), which embed the structural constraints of SCMs within deep learning frameworks to ensure that learned representations adhere to causal principles rather than mere statistical associations [102]. A critical insight driving this field is that expressivity alone is insufficient for causal learning; even arbitrarily complex neural networks cannot infer interventional effects from observational data without specific inductive biases that encode causal structure [102].

A foundational element in unifying these fields is the Principle of Independent Causal Mechanisms (ICM), which posits that the mechanism generating a cause is independent of the mechanism generating the effect given the cause. This principle serves as a powerful inductive bias for learning robust models that generalize across different environments. By leveraging distribution shifts between training environments, frameworks derived from the ICM principle enable neural networks to focus on invariant relations corresponding to true causal mechanisms while ignoring unstable, spurious correlations [289]. This approach is particularly vital for domain generalization, where standard statistical models fail due to changes in data distribution. Methods such as Causality Inspired Representation Learning (CIRL) formalize this by assuming inputs consist of both causal and non-causal factors, enforcing properties like joint independence and causal sufficiency to extract representations that remain valid in unseen target domains [214]. Furthermore, recent work suggests that many causal representation learning approaches can be unified under a broader invariance principle, where the identification of causal variables is guided by preserving data symmetries rather than strictly adhering to hierarchical causal assumptions [UNIFYING CAUSAL REPRESENT LEARNING WITH THE INVARIANCE PRINCIPLE].

The integration of graph structures has further solidified the connection between causal theory and deep learning architectures. Graph Neural Networks (GNNs), acting as universal approximators on structured inputs, have been theoretically linked to SCMs, providing a viable candidate for learning causal mechanisms from partially observable systems. Research establishes that specific classes of GNN-based models are both necessary and sufficient for causal effect identification, effectively translating the abstract requirements of causal inference into concrete neural network designs [247]. This synergy extends to causal discovery, where differentiable optimization techniques allow for the learning of causal graphs directly from data. For instance, methods like NOTEARS and its successors formulate the acyclicity constraint as a differentiable equation, enabling the use of gradient-based solvers to discover causal structures in high-dimensional spaces, a task previously hindered by combinatorial explosion [25]. Additionally, specialized architectures such as Causality-Informed Neural Networks (CINN) systematically map discovered hierarchical causal structures into layer-to-layer designs, ensuring that the orientation of causal relationships is strictly preserved during training [8].

Despite these advances, significant challenges remain regarding identifiability and the handling of latent variables. Causal Representation Learning (CRL) aims to recover high-level causal variables from low-level observations, a problem that is inherently ill-posed without additional constraints. Recent theoretical work has established sufficient and necessary conditions for identifiability in latent additive noise models by leveraging distribution shifts arising from soft interventions, extending beyond the limitations of linear or polynomial models [96]. Moreover, the concept of causal abstraction has gained traction, focusing on mapping SCMs across different levels of granularity to ensure interventional consistency between micro and macro levels [290]. This is complemented by methods like Targeted Causal Reduction, which condense complex intervenable models into concise sets of causal factors to explain specific phenomena, thereby enhancing interpretability in high-dimensional scientific simulations [208].

The unification also addresses the limitations of traditional instrumental variable (IV) methods through deep generative models. By learning representations that decompose observed covariates into confounding and instrumental components, new architectures can perform IV-based estimation even when explicit instruments are unavailable, effectively expanding the scope of causal inference in observational settings [272]. Similarly, deep structural causal models (DSCMs) utilizing normalizing flows and variational inference have enabled tractable counterfactual inference, allowing systems to answer queries at all three rungs of Pearl's ladder of causation: association, intervention, and counterfactuals [139]. However, contradictions and gaps persist, particularly concerning the "indefinite data" paradigm where causal structures and deep representations may conflict, necessitating self-supervised frameworks to enforce consistency between structural and representational learning [207]. The field continues to evolve towards more modular and composable systems, where causal knowledge can be reused and adapted across environments, moving closer to the goal of building artificial intelligence capable of robust causal reasoning [194].

Data Paradigms and the Scope of Causal Deep Learning

The integration of causal inference with deep learning has necessitated a fundamental re-evaluation of the data objects upon which causal models operate, creating a spectrum that ranges from definite statistical data to semi-definite deep learning inputs, and finally to emerging indefinite data paradigms. This stratification is critical because the transition from traditional causal inference to deep causal learning involves not merely scaling algorithms, but redefining the very objects of study-causal variables and their representations. In conventional settings, causal models operate on definite data, characterized by single-structure causal models and single-value variables, typically found in tabular statistical datasets [6]. These scenarios allow for the direct application of Structural Causal Models (SCMs) where variables are observed and relationships are often linear or parametrically constrained. However, the integration of deep learning has necessitated a shift toward semi-definite data, which encompasses complex formats such as images, text, and time-series. In these domains, low-level observations (e.g., pixels or tokens) must be transformed into high-level causal representations, introducing challenges related to latent variable identification and nonparametric mixing functions [2].

The distinction between these paradigms lies in the complexity of the causal structure and the variability of the variables. While definite data assumes a fixed structure with observed variables, semi-definite data often involves multi-value variables where the causal structure may remain single but the representation is learned through deep neural networks [6]. This shift requires methods capable of handling unstructured data, leveraging the representational power of neural networks to approximate data generation mechanisms and solve combinatorial optimization problems inherent in causal discovery [2]. For instance, causal representation learning (CRL) aims to recover latent causal variables from high-dimensional observations, a task that is ill-posed without additional assumptions such as multiple environments or specific intervention types [98]. Recent theoretical advances have shown that under general environments with nonparametric mixing, it is possible to identify latent variables and their directed acyclic graphs (DAGs) by leveraging sufficient changes in causal mechanisms, moving beyond the restrictive linearity assumptions of earlier works [98]. Furthermore, identifiability in these semi-definite contexts can be achieved by characterizing distribution shifts, where soft interventions across environments allow for the recovery of latent additive noise models up to trivial transformations [96].

Emerging from this progression is the concept of indefinite data, a paradigm characterized by multi-structure data and multi-value representations, exemplified by dialogues and video sources. This category represents a significant frontier where existing causal discovery methods often fail due to the coexistence of dynamic structures and complex representations [168]. Indefinite data induces low sample utilization and violates standard distributional assumptions, rendering traditional causal inference techniques infeasible [263]. The primary challenge in this domain is the "causal inconsistency" between the causal relationships expressed by the structure and those generated by deep learning representations, a phenomenon unique to indefinite data types [207]. To address these gaps, researchers have begun proposing probabilistic frameworks that treat causal strength as a latent variable and disentangle causal graphs into observed and latent relation subgraphs, thereby accommodating the fluid nature of indefinite data [263]. Additionally, novel baseline models incorporate causation conditions based on noise independence under non-fixed structures to handle the method gaps inherent in these multi-structure environments [168].

The interplay between these paradigms highlights a crucial tension in the field: the trade-off between identifiability and flexibility. While definite data offers strong identifiability under strict assumptions, it lacks the capacity to model real-world complexity. Conversely, indefinite data captures this complexity but suffers from severe identifiability issues without novel constraints, such as grouping observational variables or leveraging unknown interventions [297]. Furthermore, the presence of latent confounders complicates the transition across paradigms, requiring methods that can distinguish between spurious correlations and true causal mechanisms in high-dimensional spaces [14]. The development of differentiable causal discovery algorithms and neural additive models has begun to bridge this gap, allowing for the estimation of nonlinear causal relationships in time-series and other semi-definite domains [288]. However, the roadmap for indefinite data remains nascent, requiring new theoretical foundations to handle the lack of fixed structures and the dynamic nature of variables inherent in modalities like video and dialogue [6]. Understanding these distinctions is not merely academic; it is a prerequisite for selecting appropriate techniques that can robustly handle latent variables and mixing functions across the evolving landscape of data types [2].

Theoretical Foundations: Identifiability and Latent Variable Recovery

A central challenge in causal representation learning (CRL) is the problem of identifiability: determining the precise conditions under which latent causal variables and their structural relationships can be uniquely recovered from high-dimensional observed data. This problem is inherently ill-posed because it combines two difficult tasks-representation learning and causal discovery-where different latent representations can often explain the observed data equally well without additional constraints. Consequently, purely observational independent and identically distributed (i.i.d.) data is generally insufficient to identify latent causal structures in nonparametric settings, as demonstrated by the fundamental ambiguities in nonlinear Independent Component Analysis (ICA) [297]. To resolve these symmetries, theoretical advancements have established that specific assumptions regarding data structure, environmental heterogeneity, or functional forms are necessary to break the indeterminacy inherent in unsupervised learning.

One prominent strategy for achieving identifiability involves leveraging structural constraints on the observation process itself, such as multiple views or specific grouping patterns. In scenarios where data is collected from simultaneously observed views, such as different sensor modalities or biomedical measurements, it is possible to learn the information shared across these views up to a smooth bijection, even when each view constitutes a nonlinear mixture of only a subset of latent variables [173]. This framework utilizes contrastive learning and provides graphical criteria, termed "identifiability algebra," to determine which latent variables can be recovered based on the overlap of observations. Similarly, in contexts where multi-view data is unavailable, assuming that observational variables exhibit a suitable grouping structure allows for identifiability without requiring temporal structure, interventions, or weak supervision [297]. These approaches suggest that partial observability across multiple views or structured grouping can yield finer-grained representations than single-view methods, effectively relaxing the stringent requirements needed for standard nonlinear ICA.

When structural constraints on the mixing function are unavailable, researchers frequently turn to data collected across multiple environments or under interventions to provide the necessary signal for recovery. A significant body of work establishes that distribution shifts arising from interventions are critical for breaking identifiability ambiguities. While early results often relied on restrictive assumptions, such as single-node hard interventions or linear mixing functions, recent theoretical developments have substantially relaxed these constraints. For instance, identifiability has been proven for settings involving multi-node interventions within a single environment, provided there is sufficient diversity and coverage of interventions across domains [196]. Furthermore, in the general nonparametric setting where both the causal model and mixing function are unknown, it has been shown that observational data combined with perfect interventions on each node suffice for identifiability, subject to genericity conditions that rule out fine-tuned solutions [252]. Extending this to general environments, including soft interventions and nonstationary time series, recent work demonstrates that leveraging sufficient change conditions on causal mechanisms up to third-order derivatives allows for the full recovery of the latent Directed Acyclic Graph (DAG) and latent variables up to minor indeterminacies [98]. This represents a significant advancement over prior works that could only recover the moral graph or identify variables up to ancestral relationships under similar nonparametric conditions [35].

In the absence of explicit intervention targets or multi-view data, sparsity and specific inductive biases offer alternative pathways to identifiability. The principle of mechanism sparsity regularization posits that if latent factors depend sparsely on past factors or auxiliary variables, disentangled representations can be recovered by simultaneously learning the latent factors and the sparse causal graph [123]. This approach connects nonlinear ICA with causality, showing that regularizing for sparsity can resolve ambiguities that persist under standard independence assumptions. Similarly, assuming a single-parent decoding structure, where each observed low-level variable is affected by only one latent variable, enables the identification of latent causal processes in temporal data [113]. Additionally, the use of additive noise models (ANM) as an inductive bias has been shown to restrict admissible transformations to the affine class, thereby resolving component-wise indeterminacy in unsupervised settings where general diffeomorphisms would otherwise prevent unique recovery [130].

Despite the diversity of these approaches, a unifying perspective suggests that many identifiability results rely on aligning representations with inherent data symmetries rather than strictly adhering to causal hierarchies. The invariance principle serves as a guiding mechanism, where causal variables are identified because they remain invariant across specific transformations or environments, regardless of whether those transformations are strictly causal in nature [209]. This perspective allows for the mixing of causal and non-causal assumptions based on the relevant invariances of the problem domain. However, contradictions and limitations remain; for example, while some methods achieve full DAG recovery under nonparametric mixing with general environments, others may only recover the moral graph depending on the strength of the intervention and the linearity of the mixing function [35]. Furthermore, in temporal settings, the presence of instantaneous effects complicates identifiability, requiring specialized frameworks that can handle both time-delayed and instantaneous causal relations without relying on overly simplistic Markov assumptions [157]. Ultimately, the consensus indicates that while purely observational i.i.d. data is insufficient, the strategic incorporation of environmental heterogeneity, sparsity, multi-view structures, or specific noise models provides a robust theoretical foundation for recovering latent causal variables.

Identifiability via Interventions and Environmental Heterogeneity

The fundamental challenge in causal representation learning lies in the fact that inferring high-level causal variables from low-level observations is inherently underconstrained when relying solely on observational data. While early theoretical frameworks often depended on restrictive assumptions, such as the requirement that interventions target only a single variable within any given environment, recent advancements have significantly relaxed these constraints. A pivotal shift has been the move toward allowing multiple variables to be targeted simultaneously within one environment. This approach leverages the specific trace that interventions leave on the variance of ground truth causal variables, regularizing for sparsity with respect to this trace to recover latent structures [196]. Furthermore, identifiability guarantees have been established for scenarios involving unpaired observational and interventional data, where soft interventions alter the mechanism of latent variables. Under a generalized notion of faithfulness, these methods ensure the recovery of the latent causal model up to an equivalence class, even when causal variables are unobserved [224].

The scope of these identifiability results has expanded beyond independent and identically distributed (i.i.d.) data settings to include structured contexts, such as network data. Frameworks like GRACE-VAE integrate discrepancy-based variational autoencoders with graph neural networks, demonstrating that theoretical identifiability results derived from i.i.d. data hold even in non-i.i.d. settings where structured context is available [12]. In the most general nonparametric settings, where both the causal model and the mixing function are unknown and unrestricted, research indicates that the observational distribution combined with perfect interventions per node suffices for identifiability, subject to genericity conditions that rule out fine-tuned spurious solutions. For an arbitrary number of variables, the presence of at least one pair of distinct perfect interventional domains per node guarantees the identification of both ground truth latents and their causal graph [252].

While early breakthroughs in leveraging environmental changes for identifiability were limited to linear Gaussian models, recent developments have extended this to nonlinear causal relationships represented by polynomial models and general noise distributions conforming to the exponential family. These studies investigate the necessity of imposing changes on all causal parameters and provide partial identifiability results when only a subset of parameters changes, validating these findings through empirical estimation methods [271]. Pushing the boundaries further, current research formalizes desiderata for causal representation learning from "general environments," which encompass a broader class of distribution changes including soft and coupled interventions. By leveraging sufficient change conditions on causal mechanisms up to third-order derivatives, it is now possible to fully recover the latent Directed Acyclic Graph (DAG) and identify latent variables up to minor indeterminacies under nonparametric mixing and nonlinear latent causal models, such as additive or heteroscedastic noise models [98]. This represents a significant improvement over prior works that could only identify the moral graph or required linear mixing functions.

The nature of the distribution shifts themselves is critical for identifiability. Theoretical work has established sufficient and necessary conditions characterizing the types of distribution shifts that contribute to identifiability within latent additive noise models. These findings extend beyond linear and polynomial models, offering flexibility through Multilayer Perceptrons while addressing issues of numerical instability. Moreover, partial identifiability results are provided for practical scenarios where only a portion of the data exhibits the required distribution shifts [96]. In temporal settings, frameworks have been developed to recover time-delayed latent causal variables and identify their relations under different distribution shifts. These approaches factorize unknown shifts into transition distribution changes caused by fixed dynamics versus time-varying latent causal relations, enabling efficient model correction with few samples from new environments [264]. Similarly, in nonstationary settings, recent studies show that independent latent components can be recovered from nonlinear mixtures without observing auxiliary variables, provided mild conditions on the Markov assumption are met [219].

Contradictions and refinements exist regarding the strength of assumptions needed. While some approaches rely on hard interventions to enable nonparametric models, others argue that soft interventions align better with real-world self-initiated behaviors and offer greater flexibility, though they may require specific conditions on the derivatives of causal mechanisms [96]. Additionally, the role of invariance is being re-evaluated; some works suggest that many causal representation learning approaches methodologically align representations with inherent data symmetries guided by invariance principles that are not necessarily causal, unifying various approaches under a single method that can mix causal and non-causal assumptions [209]. Despite these advances, challenges remain in fully identifying causal structures without any interventions, leading to alternative strategies such as leveraging grouping of observational variables or mechanism sparsity regularization to achieve identifiability in weakly supervised or unsupervised settings [297] [123]. The integration of these diverse theoretical strands highlights a consensus that environmental heterogeneity and interventional data are pivotal for disentangling latent causal structures, though the specific requirements regarding the number of environments, the nature of interventions, and the functional forms of causal mechanisms continue to be refined.

Linear vs. Nonlinear and Temporal Identifiability

The trajectory of causal representation learning has undergone a paradigm shift, moving from restrictive linear assumptions reliant on non-Gaussian noise toward flexible nonlinear frameworks capable of capturing the complexity of real-world dynamical systems. Early foundational work primarily established identifiability for linear structural causal models (SCMs) by leveraging non-Gaussianity to ensure the unique recovery of latent variables. However, the assumption of linearity often fails to capture the intricate, nonlinear dynamics inherent in natural systems. To address this limitation, recent theoretical advances have extended identifiability guarantees to general nonlinear settings under progressively weaker assumptions. A pivotal development in this evolution involved leveraging distribution shifts across multiple environments to identify latent causal structures without the necessity of hard interventions. While initial breakthroughs utilized changes in causal influences within linear Gaussian models, subsequent research has generalized these findings to encompass nonlinear causal relationships represented by polynomial models and general noise distributions conforming to the exponential family [271]. Although polynomial models possess universal approximation capabilities, they are frequently plagued by numerical instability and an exponential growth in terms when high-degree approximations are required, limiting their practical utility in high-dimensional settings.

To overcome the limitations inherent in polynomial constraints, researchers have proposed latent additive noise models implemented through Multilayer Perceptrons (MLPs), which offer superior flexibility while maintaining rigorous identifiability conditions. It has been established that latent additive noise causal models can be identified up to trivial permutation and scaling transformations, provided there is a sufficient and necessary condition characterizing the types of distribution shifts present in the data [96]. This framework is significant because it unifies previous disjoint results, encompassing both latent linear models and latent polynomial models as special cases. Furthermore, the theory extends to latent post-nonlinear causal models, a more general class that includes additive noise models as a subset. In scenarios where data availability is limited and only a portion of the data exhibits the required distribution shifts, partial identifiability results have been derived, effectively narrowing the gap between theoretical ideals and practical constraints [96]. Complementing this, other works have demonstrated that even in the absence of temporal structure or explicit interventions, identifiability can be achieved by assuming a suitable grouping of observational variables, which resolves component-wise indeterminacy by restricting admissible transformations to the affine class [297].

The transition from linear to nonlinear identifiability also necessitates a re-evaluation of the nature of interventions and confounding. While hard interventions enable nonparametric latent causal models, they often require specific, somewhat arbitrary distribution shifts that may not align with self-initiated behaviors in causal systems. In contrast, soft interventions align with a wider range of potential distribution shifts and have been shown to facilitate identifiability in latent additive noise models [96]. Further relaxing assumptions, recent studies have proven that observational distributions combined with perfect interventions per node suffice for identifiability in general nonparametric settings, subject to genericity conditions that rule out spurious solutions [252]. Additionally, in the context of time series, the structural equation process (SEP) representation allows for the generalization of linear Gaussian SCM concepts, such as Wright's path-rule and the trek-rule, to structural vector autoregressive (SVAR) processes at the level of process graphs [193]. This enables the identification of causal effects in the frequency domain without needing to resolve the full complexity of the infinite time series graph, provided the process graph structure is known or identifiable.

In the temporal domain, the challenge of identifiability becomes increasingly complex due to the presence of time-delayed and instantaneous dependencies. Traditional methods often assume that latent causal processes lack instantaneous relations or rely on stationary environments, which limits their applicability to dynamic real-world systems. Recent frameworks have addressed this by establishing identifiability theories for nonparametric latent causal dynamics from their nonlinear mixtures, even under fixed dynamics and varying distribution shifts [264]. These approaches factorize unknown distribution shifts into transition distribution changes caused by fixed dynamics and time-varying latent causal relations, allowing for the reliable identification of time-delayed latent causal influences. Specifically, the LiLY framework and related temporally disentangled representation learning methods demonstrate that time-delayed latent causal variables can be recovered from measured sequential data by exploiting modular representations of changes [99]. Moreover, new identification frameworks have been proposed to handle instantaneous latent dynamics by imposing sparse influence constraints, establishing identifiability up to a Markov equivalence class based on sufficient variability and contextual information [157].

Despite these significant advances, contradictions and gaps remain regarding the strength of assumptions required for full versus partial identifiability. For instance, while some methods guarantee unique recovery under strong assumptions about the diversity of interventions across environments, others provide only equivalence class recovery under milder conditions [224]. Furthermore, the assumption of additive noise, while powerful, does not guarantee unique identifiability in the general mixing case without additional inductive biases or auxiliary signals [130]. The integration of neural networks into these frameworks, such as in Differentiable Invariant Causal Discovery (DICD), utilizes multi-environment information to discover environment-invariant causation, theoretically guaranteeing identifiability under mild conditions given enough environments [248]. Collectively, these developments illustrate a robust trajectory toward identifying complex, nonlinear, and temporal causal structures from observational and interventional data, moving well beyond the constraints of early linear Gaussian models.

The Role of Mixing Functions and Observation Models

The central obstacle in causal representation learning (CRL) is the inherent ambiguity introduced by the mixing function, which maps low-dimensional latent causal variables to high-dimensional observations. Historically, recovering the latent Directed Acyclic Graph (DAG) and the underlying variables required restrictive parametric assumptions, such as linearity in the mixing function or specific distributional constraints on the latents. However, recent theoretical breakthroughs have demonstrated that full recovery is achievable even under nonparametric mixing, provided that specific conditions regarding environmental heterogeneity, intervention types, or temporal structures are met. This paradigm shift challenges the necessity of strong parametric priors, suggesting instead that the diversity of data generation processes can sufficiently constrain the solution space.

A foundational result in this domain establishes that in a general setting where both the causal model and the mixing function are nonparametric, identifiability can be achieved using multiple datasets arising from unknown interventions [252]. For the fundamental case of two causal variables, the authors prove that an observational distribution combined with one perfect intervention per node suffices for identifiability, subject to a genericity condition that precludes spurious solutions involving fine-tuning. This requirement extends to arbitrary numbers of variables, where at least one pair of distinct perfect interventional domains per node guarantees recovery. Crucially, this work demonstrates that the strengths of causal influences are preserved across all equivalent solutions, ensuring the inferred representation remains robust for causal conclusions. This stands in contrast to earlier approaches that relied on weak supervision via counterfactual views or imposed linearity on the mixing function.

Expanding beyond hard interventions, which are often unrealistic in fields like genomics, recent work formalizes a broader class of "general environments" to address soft interventions [98]. This research shows that one can fully recover the latent DAG and identify latent variables up to minor indeterminacies under nonparametric mixing and nonlinear latent causal models, such as additive or heteroscedastic noise models. The key theoretical innovation involves leveraging sufficient change conditions on causal mechanisms up to third-order derivatives. While previous studies could only identify the moral graph of the latent DAG using second-order derivatives, the utilization of third-order derivatives allows for the extraction of causal ordering information, enabling full DAG recovery [98]. This result matches or improves upon existing works while requiring less restrictive assumptions about how distributions change across environments.

The nature of interventions further influences the degree of identifiability. In scenarios where unpaired observational and interventional data are available, identifiability can be achieved with unobserved causal variables given a generalized notion of faithfulness [224]. This approach guarantees the recovery of the latent causal model up to an equivalence class, which is particularly relevant for predicting the effects of unseen combinations of interventions. Similarly, investigations into latent additive noise models establish a sufficient and necessary condition characterizing the types of distribution shifts suitable for identification [96]. By extending beyond linear or polynomial models, these findings demonstrate that latent additive noise causal models can be identified up to trivial permutation and scaling transformations, provided the distribution shifts meet specific criteria regarding the exponential family of noise distributions.

Temporal dynamics offer another powerful avenue for achieving identifiability without strict parametric mixing assumptions. Theoretical frameworks have been established for nonparametric latent causal processes from their nonlinear mixtures under fixed temporal causal influences [99]. The proposed TDRL framework factors unknown distribution shifts into transition distribution changes, reliably identifying time-delayed latent causal influences. This approach outperforms baselines that fail to exploit the modular representation of changes. Furthermore, the LiLY framework recovers time-delayed latent causal variables under different distribution shifts by factorizing shifts into transition distribution changes caused by fixed dynamics and time-varying relations [264]. These temporal methods highlight that the structure of time itself can serve as a potent constraint for disentangling nonparametric mixtures, effectively replacing the need for diverse interventional environments with sequential information.

Despite these advances, contradictions and limitations remain regarding the extent of identifiability under varying conditions. While some works prove full recovery under perfect interventions, others suggest that under softer conditions or partial observability, recovery may be limited to an equivalence class or specific subspaces. Addressing this, a unified framework for multi-view CRL presents that latent variables can be identified up to a smooth bijection using contrastive learning across multiple views, suggesting that partial observability can be mitigated by multi-modal data rather than just interventional diversity [173]. Additionally, the concept of "partial disentanglement" argues that complete identifiability may not always be achievable or necessary; instead, a qualitative understanding of which factors remain entangled based on graph sparsity can be sufficient for many applications [59].

The interplay between the mixing function and the observation model is also critical in hierarchical and continuous-time settings. Recent proposals suggest that the joint distribution of hierarchical latent variables can be uniquely determined using three conditionally independent observations, leveraging natural sparsity to identify variables within each layer [38]. Meanwhile, extensions to continuous-time stochastic point processes analyze the geometry of the parameter space to address gaps left by methods assuming i.i.d. or discrete-time processes [81]. These diverse approaches collectively indicate that while the mixing function introduces significant ambiguity, the strategic exploitation of environmental heterogeneity, temporal structure, and higher-order derivative information can effectively constrain the solution space, allowing for the recovery of meaningful causal representations even in the absence of strong parametric assumptions.

Factorized Representations in Biological and Multimodal Systems

The application of identifiability theory to biomedical data has catalyzed a paradigm shift from purely predictive modeling toward the recovery of latent causal structures that govern cellular and physiological responses. A central challenge in single-cell genomics and multimodal biomedical analysis is the disentanglement of mixed signals arising from distinct biological sources, such as genetic background, environmental perturbations, and their complex interactions. Recent methodological advances have successfully addressed this by proposing factorized representations that explicitly separate covariate-specific, treatment-specific, and interaction-specific components. Notably, the Factorized Causal Representation (FCR) learning method demonstrates that by leveraging the framework of identifiable deep generative models, it is possible to learn disentangled cellular representations comprising covariate-specific ($\mathbf{z}_x$), treatment-specific ($\mathbf{z}_t$), and interaction-specific ($\mathbf{z}_{tx}$) blocks [31]. This approach provides rigorous theoretical guarantees, proving component-wise identifiability for interaction terms and block-wise identifiability for covariate and treatment terms, thereby outperforming state-of-the-art baselines in predicting cellular responses across diverse cell lines [31].

The theoretical underpinnings of these factorized models often rely on the assumption of sparse mechanism shifts, positing that perturbations target only a sparse subset of latent variables rather than altering the entire generative process. This principle is formalized in methods that treat genetic or chemical perturbations as stochastic interventions on unknown latent factors, enabling the recovery of causal semantics from non-stationary data [183]. By extending nonlinear Independent Component Analysis (ICA) theory to biological contexts, researchers have shown that such sparsity constraints, combined with interventional data, allow for the identification of latent causal models even when causal variables are unobserved [224]. Furthermore, the integration of temporal dynamics into these frameworks addresses the limitation of static models, proposing latent dynamical causal generative models that jointly capture cellular programs and their evolution over time, ensuring recoverability up to a trivial equivalence class under suitable conditions [105].

In the realm of multimodal biomedical observations, where data arises from heterogeneous sources such as imaging and sequencing, identifiability becomes increasingly complex due to the potential for restrictive parametric assumptions in existing models. To overcome this, recent work establishes flexible identification conditions for nonparametric latent distributions, utilizing the structural sparsity of causal connections between modalities as a natural inductive bias for biomedical systems [300]. This aligns with unified frameworks for multi-view causal representation learning, which prove that information shared across partially observed views can be learned up to a smooth bijection, effectively refining the granularity of identified representations under milder observability assumptions [173]. These multimodal approaches are critical for human phenotype research, where understanding the interplay between different data types is essential for elucidating underlying physiological mechanisms.

Despite these advances, contradictions and limitations persist regarding the strength of identifiability guarantees under varying data conditions. While some studies assert complete disentanglement under specific graph connectivity criteria and sparsity regularization [123], others argue that without strict adherence to these criteria, only "partial disentanglement" is achievable, where certain factors remain entangled based on the ground-truth graph structure [59]. This nuance is further explored in nonparametric settings, where identifiability is shown to hold up to a novel equivalence relation called consistency, acknowledging that full separation of factors may not always be possible without stronger supervision or specific intervention patterns [55]. Additionally, the presence of confounders poses a significant barrier; standard assumptions of unconfoundedness often fail in biological systems, necessitating frameworks like Confounded-Disentanglement (C-Disentanglement) that explicitly introduce inductive biases via domain expertise to identify causally disentangled factors [172].

The practical utility of these factorized representations extends to improving robustness against distribution shifts and enhancing interpretability in clinical settings. Methods leveraging task structures and conditional priors have demonstrated that linear identifiability can be achieved in multi-task regression settings, reducing the equivalence class to permutations and scaling, which is crucial for recovering canonical representations in molecular data [255]. Moreover, the integration of causal representation learning with survival analysis has yielded frameworks like TV-SurvCaus, which synergize representation balancing with sequential modeling to estimate causal effects of time-varying treatments while addressing treatment-confounder feedback [237]. However, a fundamental tension remains between predictive accuracy and causal fidelity; recent impossibility theorems suggest that purely predictive objectives may structurally favor environment-dominant representations over system-specific causal factors, highlighting the necessity of explicit causal constraints to avoid the "predictive-causal gap" [91]. Consequently, the field is moving toward hybrid approaches that combine mechanistic sparsity, interventional data, and multimodal views to ensure that learned representations are not only predictive but also causally meaningful and interpretable for therapeutic discovery.

Algorithmic Advances in Causal Discovery and Structure Learning

The landscape of causal discovery has undergone a paradigm shift, moving from traditional constraint-based and score-based methods, which often struggle with combinatorial complexity and scalability, toward deep learning frameworks that recast discrete graph search into continuous optimization problems. This transition enables the inference of causal structures in high-dimensional and nonlinear regimes where classical algorithms like PC or GES become computationally prohibitive. A foundational advancement in this domain is the reformulation of the acyclicity constraint as a differentiable equation, allowing for gradient-based optimization of causal graphs. Building on this, recent neural architectures have integrated Graph Neural Networks (GNNs) to capture complex structural dependencies. For instance, [195] introduces a probabilistic framework that learns a distribution over the space of causal graphs rather than outputting a single deterministic structure. By encoding both node and edge attributes, this approach leverages global structural information and statistical measures like mutual information to outperform deterministic GNN-based methods and traditional algorithms on both synthetic and real-world datasets. Theoretical underpinnings for such integration are further solidified in [247], which establishes necessary and sufficient conditions for GNN-based models to identify causal effects, bridging the gap between universal approximation capabilities and structural causal modeling.

Scalability remains a critical challenge as the number of variables increases to the thousands, particularly in genomics and neuroscience. To address the memory and time bottlenecks associated with attention mechanisms in large graphs, [45] proposes a neural architecture featuring a reduction unit for compressing data embeddings and tied attention weights to improve space efficiency. This two-stream design, which separates data evidence extraction from statistical graph priors, successfully scales inference to graphs with up to 1,000 nodes, achieving significant speedups over prior methods. Similarly, [56] tackles high-dimensional interventional data by introducing factor directed acyclic graphs (f-DAGs). By restricting the search space to non-linear low-rank causal interaction models, this method achieves causal discovery on thousands of variables, demonstrating superior performance in large-scale single-cell RNA sequencing datasets compared to state-of-the-art baselines. These methods collectively demonstrate that by combining architectural efficiency with structural priors, deep learning can overcome the scalability limits that previously restricted causal discovery to small-scale systems.

The application of these advances extends significantly into temporal domains, where dynamic causality and irregular sampling present unique hurdles. Moving beyond linear Vector AutoRegression (VAR) models, [288] generalizes the framework to an additive nonlinear setting, using deep neural networks to extract Granger causal influences while maintaining interpretability. Addressing the complexities of irregular sampling and missing data, [120] utilizes a coarse-to-fine discovery technique combined with message-passing GNNs to improve scalability and robustness. Furthermore, [245] employs uncoupler and recoupler networks to disentangle time series into semantic representations, effectively capturing evolving temporal causality in systems like human motion. In spatiotemporal contexts, [217] integrates causal discovery directly into Temporal Graph Convolutional Neural Networks, using spatial and temporal decomposition to improve forecasting performance without requiring a priori domain knowledge. These temporal methods highlight the versatility of neural approaches in handling non-stationary and irregularly sampled data, a domain where traditional methods often fail.

Beyond structural learning, there is a growing emphasis on integrating discovered causal knowledge into neural network architectures to enhance robustness and generalization. [8] proposes the Causality-Informed Neural Network (CINN), which maps discovered hierarchical causal structures directly into the layer-to-layer design of the network, mitigating gradient interference and improving prediction performance. Similarly, [34] regularizes neural networks by jointly learning the causal DAG as an adjacency matrix embedded in the input layers, facilitating the discovery of optimal predictors and improving out-of-sample generalization. The need for human-in-the-loop validation is addressed by [190], which enables experts to inspect, modify, and re-inject causal graphs into neural networks, fostering contestability and debugging capabilities. Finally, emerging generative approaches such as [137] convert the graph search problem into a sequential generation task, learning policies to generate high-reward DAGs, while [246] utilizes denoising score matching to stabilize convergence in high-dimensional settings. These diverse methodological strides collectively demonstrate the transformative potential of deep learning in overcoming the limitations of traditional causal discovery, enabling scalable, robust, and interpretable inference in complex real-world systems.

Continuous Optimization and Differentiable DAG Learning

The transition from discrete combinatorial search to continuous optimization represents a paradigm shift in causal structure learning, fundamentally altering the scalability and applicability of Directed Acyclic Graph (DAG) discovery. Traditional score-based and constraint-based algorithms often encounter exponential computational complexity as variable counts rise, rendering them impractical for the high-dimensional datasets prevalent in biomedicine and genomics [25]. By recasting the discrete graph search as a continuous optimization problem solvable via standard deep learning optimizers, researchers have bypassed the need for exhaustive search strategies. While early formulations relied on trace exponential constraints to enforce acyclicity, recent advancements have refined these mechanisms to handle non-parametric settings and improve numerical stability. For instance, [176] utilizes Reproducing Kernel Hilbert Spaces (RKHS) combined with a log-determinant formulation for the acyclicity constraint. This approach ensures stability and effectively models complex, non-linear relationships without relying on finite-dimensional approximations, addressing the limitations of linear structural equation models that dominate earlier continuous approaches. Furthermore, to mitigate stability issues in high-dimensional, feature-imbalanced data where standard methods like NOTEARS often struggle, [246] proposes using denoising score matching objectives to smooth the optimization landscape. This method, coupled with an adaptive k-hop acyclicity constraint, improves runtime by avoiding costly matrix inversions, demonstrating that the choice of optimization objective is as critical as the acyclicity formulation itself.

This continuous framework has proven particularly transformative for dynamic causal discovery in time-series data, where relationships evolve over time and computational demands often grow exponentially. Moving away from the computationally expensive matrix exponential constraints used in earlier dynamic Bayesian network approaches, [198] introduces a constraint-free method utilizing a quasi-maximum likelihood score function. LOCAL employs adaptive modules for causal mask learning and dynamic graph parameterization to efficiently recover dynamic DAGs in high-dimensional settings, effectively decoupling the algebraic characterization of acyclicity from the optimization loop. Similarly, [245] leverages uncoupler and recoupler networks to disentangle time series into semantic representations, estimating dynamic causal influences through datapoint-wise prediction errors. These methods are complemented by approaches like [120], which addresses the challenge of missing entries and irregular sampling through a coarse-to-fine discovery strategy and message-passing graph neural networks. Additionally, [43] reinforces this trend by framing dynamic Bayesian network inference as a link prediction problem via Graph Neural Networks, significantly outperforming linear structural causal model baselines. Together, these works illustrate the versatility of continuous optimization in capturing temporal dependencies that static models frequently miss.

Despite the scalability benefits, differentiable methods face persistent challenges regarding identifiability, particularly in the presence of latent confounders and distribution shifts. Purely observational differentiable learning can converge to spurious solutions due to data biases. [248] addresses this by utilizing multi-environment data to discover environment-invariant causation, effectively filtering out spurious correlations that vary across domains. This aligns with theoretical advances in [98], which demonstrates that latent DAGs can be fully recovered under non-parametric mixing functions by leveraging sufficient change conditions up to third-order derivatives. In scenarios involving unmeasured confounders, [279] develops a gradient-based approach for learning Acyclic Directed Mixed Graphs (ADMGs) with non-linear functional relations, overcoming the tractability issues of discrete search. Additionally, to manage the vast search space in ultra-high-dimensional regimes, [56] introduces factor DAGs (f-DAGs) to restrict the search to low-rank causal interaction models, enabling discovery with thousands of variables.

The flexibility of differentiable frameworks also facilitates the integration of prior knowledge and hybrid architectures to guide the optimization process. Recognizing that purely data-driven methods may suffer from a cold-start problem or lack interpretability, [229] proposes using Large Language Models to initialize the adjacency matrix optimization. This provides strong priors that guide the gradient descent process toward more accurate and interpretable causal graphs. Similarly, [71] emphasizes that while knowledge becomes less critical with massive data, it remains vital for reducing search space complexity in data-scarce regimes. Conversely, challenging the necessity of non-convex optimization entirely, [158] suggests a decoupled paradigm that extracts topological order directly from score functions via algebraic procedures, reframing the problem as one of statistical estimation rather than constrained optimization. These diverging paths illustrate the ongoing evolution of the field, balancing the trade-offs between computational efficiency, model flexibility, and theoretical identifiability to expand the applicability of causal discovery to complex, real-world systems.

Integrating Domain Knowledge and Large Language Models

The integration of domain knowledge and Large Language Models (LLMs) into causal discovery frameworks represents a pivotal algorithmic advancement designed to overcome the scalability limitations, local optima convergence, and cold-start deficiencies inherent in purely data-driven approaches. Traditional differentiable causal discovery methods, while powerful in extracting relationships from observational data, frequently lack interpretability and struggle to incorporate specific prior knowledge without rigid constraints. To bridge this gap, recent research has shifted towards hybrid architectures that leverage the semantic reasoning capabilities of LLMs to initialize or guide the optimization processes of structural learning algorithms. A foundational strategy involves using LLMs to generate strong priors that inform the maximum likelihood objective functions. For instance, the [229] framework explicitly defines the adjacency matrix as the sole variational parameter, allowing an LLM to initialize the optimization process. This approach not only enhances interpretability by making the graph structure explicit but also provides empirical evidence that the quality of the LLM-generated initialization directly correlates with the accuracy of the final causal graph, effectively merging formal causal reasoning with semantic knowledge.

Building upon static initialization, more sophisticated frameworks have emerged to dynamically balance data-driven evidence with semantic priors throughout the learning trajectory, moving beyond fixed starting points. The [256] framework exemplifies this evolution by employing a dual-encoder architecture that integrates LLM-generated adjacency matrices with observational data, utilizing a reinforcement learning agent to optimize both computational efficiency and accuracy simultaneously. Unlike methods relying on fixed priors, GUIDE dynamically balances reward maximization against Directed Acyclic Graph constraints, enabling robust performance across mixed data types and scalability to graphs with 70 or more nodes-a regime where traditional algorithms often fail due to prohibitive energy costs. This dynamic balancing act is critical, as research indicates that while reduced search spaces derived from knowledge can improve accuracy, they do not always guarantee reduced computational complexity, particularly when the relationships implied by the data and the injected knowledge are in tension [71]. Consequently, the most effective algorithms are those that allow for a flexible interplay between data-driven learning and knowledge injection rather than imposing rigid constraints that may conflict with observational evidence.

Beyond initialization and dual-encoding, iterative supervision mechanisms have been developed to refine causal structures through continuous feedback loops, addressing the issue of unreliable constraints from imperfect LLM inferences. The [180] framework integrates LLM-based causal inference into an iterative refinement process, generating robust structural constraints by leveraging feedback to progressively improve the causal DAG. Similarly, the [253] framework extends the utility of LLMs to the discovery of hidden variables within unstructured data. Instead of directly inferring causality, this approach constructs feedback from intermediate causal discovery results to prompt the LLM to propose and refine high-level measured variables, effectively uncovering Markov Blankets and Partial Ancestral Graphs in complex domains. These approaches collectively suggest that LLMs are most effective when used to constrain the search space or propose variables rather than acting as standalone causal reasoners, a finding supported by studies showing that pairing LLMs with causal machine learning algorithms produces graphical structures that align more closely with domain expert knowledge than those derived from causal ML alone [74].

The versatility of these integrated methods is further validated in specialized scientific domains where ground truth is often unavailable or data is sparse. In biological research, the [115] framework leverages the learned biological knowledge of LLMs to guide the discovery of Gene Regulatory Networks, using LLM-suggested networks to generate synthetic data for evaluation in scenarios lacking explicit causal graphs. Similarly, in soil science, the [226] framework combines observed data with process-based models to learn causal structures that are more representative of ground truth, subsequently improving prediction performance in out-of-sample settings. To ensure these models remain trustworthy and aligned with human expertise, recent developments in contestable neural networks enable human experts to inspect, modify, and re-inject causal graphs into the learning loop. This methodology allows practitioners to debug networks based on the causal structure discovered from the data, ensuring that the final models adhere to expert knowledge while maintaining predictive performance [190]. Furthermore, the injection of expertise logic into graph representation learning has been shown to improve model performance by ensuring that Graph Neural Networks acquire human expertise alongside end-to-end learning from datasets, reinforcing the necessity of synthesizing LLM semantics, domain expertise, and rigorous statistical learning [103].

Discovery in High-Dimensional, Biological, and Indefinite Data

The intersection of deep learning and causal discovery has become indispensable for analyzing high-dimensional biological datasets, where the sheer number of variables-often reaching into the thousands in genomics and single-cell sequencing-renders traditional combinatorial search strategies computationally intractable. Classical constraint-based and score-based algorithms frequently falter under the weight of non-linear interactions and the exponential growth of the search space. To overcome these limitations, recent advancements have introduced deep neural architectures that combine convolutional neural networks (CNNs) and graph neural networks (GNNs) within a causal risk framework. These hybrid models leverage the local feature extraction capabilities of CNNs and the relational modeling strengths of GNNs to approximate structural causal models (SCMs) at a scale previously unachievable, successfully learning causal networks from large-scale molecular profiles where conventional methods fail [7]. By transforming input vectors into image-like representations or utilizing permutation-equivariant models, these frameworks efficiently handle very large datasets, bridging the gap between universal approximation theory and causal identifiability in complex biological simulations [14] [247].

A pivotal evolution in this domain is the transition from deterministic graph learning, which outputs a single Directed Acyclic Graph (DAG), to probabilistic frameworks that quantify the inherent uncertainty in causal structures. Traditional approaches often overlook alternative valid structures supported by noisy biological data, leading to overconfident and potentially spurious conclusions. In contrast, novel GNN-based probabilistic prediction frameworks learn a distribution over the entire space of causal graphs, encoding both node and edge attributes to capture global structural information [195]. These models are trained on diverse synthetic datasets augmented with information-theoretic measures, such as mutual information and conditional entropy, enabling them to generalize to unseen real-world biological datasets without retraining. This probabilistic approach is particularly critical for indefinite data settings, where missing values and unobserved confounders are prevalent. Addressing the challenge of missing data specifically, deep learning frameworks like Imputated Causal Learning (ICL) perform iterative imputation and structure discovery simultaneously, outperforming state-of-the-art methods that treat these tasks as separate problems [24]. Furthermore, methods such as DeFuSE explicitly model unobserved confounders using deconfounding adjustments and sequential estimation procedures, demonstrating favorable performance in gene regulatory network analysis where confounding is a major concern [191].

In the specific context of biomedicine, the integration of causal inference with graph representations has proven essential for distinguishing genuine causal drivers from spurious correlations, a common pitfall in biomarker discovery. Causal Graph Neural Networks (Causal-GNN) have been developed to identify stable disease biomarkers by incorporating causal effect estimation and propensity scoring mechanisms that leverage cross-gene regulatory networks [149]. Unlike traditional feature selection methods that yield unstable biomarkers across different cohorts, these models achieve consistently high predictive accuracy by isolating invariant causal mechanisms. The scale of modern biological data has further necessitated architectures capable of handling thousands of variables through structural constraints. Innovations such as factor directed acyclic graphs (f-DAGs) restrict the search space to non-linear low-rank interaction models, enabling causal discovery on massive single-cell RNA sequencing datasets [56]. Similarly, CauScale addresses memory bottlenecks in attention-based models by compressing data embeddings and utilizing tied attention weights, successfully scaling inference to graphs with up to 1,000 nodes [45]. The emergence of Large Language Models (LLMs) further augments this capability, where LLMs are used to guide causal synthetic data generation and validate gene regulatory networks, leveraging pre-trained biological knowledge to overcome the lack of ground truth [115].

The challenge of temporal dynamics in biological and physical systems introduces additional complexity, requiring methods that can disentangle time-lagged and contemporaneous effects without relying on a priori domain knowledge. Causal Temporal Graph Convolutional Neural Networks (CTGCN) have been introduced to automatically discover underlying causal processes in spatiotemporal data, decomposing the identification problem temporally and spatially to improve forecasting performance and explainability in large-scale applications [217]. Addressing the limitations of existing methods in high-dimensional and irregularly sampled time series, CUTS+ employs a coarse-to-fine discovery strategy combined with message-passing GNNs to significantly improve performance on irregular data [120]. Moreover, UnCLe introduces a unique capability to capture evolving temporal causality in non-linear systems by disentangling input time series into semantic representations, offering a distinct advantage over static causal discovery benchmarks [245].

Despite these algorithmic strides, tensions remain regarding the handling of latent confounders and the integration of expert knowledge. While some methods assume causal sufficiency, others explicitly model unobserved confounders, highlighting a divergence in assumptions that affects identifiability. The role of domain knowledge is also contentious; while purely data-driven approaches aim to avoid bias, frameworks like CASTLE and contestable neural networks argue that injecting expert knowledge through causal graphs improves predictive performance and model parsimony [190] [34]. This hybrid approach allows practitioners to debug networks based on discovered causal structures, ensuring that models adhere to biological priors while remaining flexible enough to uncover novel mechanisms. Ultimately, the convergence of scalable architectures, probabilistic modeling, and domain-informed constraints suggests that deep learning is not merely an approximation tool but a necessary evolution for causal discovery in the era of big biological data, offering the flexibility to model complex generative mechanisms that parametric models cannot capture [77].

Conclusion

The synthesis of causal inference and deep learning represents a definitive paradigm shift from passive pattern recognition to active mechanistic understanding. By embedding structural constraints and inductive biases into flexible neural architectures, the field has moved beyond the limitations of associational learning, which frequently fails under distribution shifts due to its reliance on spurious correlations. The integration of frameworks such as Structural Causal Models with deep generative capabilities has enabled systems to ascend Pearl's ladder of causation, supporting not only prediction but also intervention and counterfactual reasoning. This evolution addresses the fundamental fragility of standard deep learning in non-i.i.d. settings, offering a pathway toward models that capture invariant causal mechanisms rather than environment-specific noise.

A central theme emerging from the literature is the resolution of the identifiability problem through environmental heterogeneity and structural sparsity. While purely observational data often renders causal discovery ill-posed, theoretical advancements have demonstrated that leveraging multi-environment data, multi-view observations, and specific interventional signals can break symmetries and ensure the unique recovery of latent causal variables. The field has successfully transitioned from restrictive linear assumptions to general nonlinear and temporal frameworks, allowing for the disentanglement of complex dynamics in biological, climate, and social systems. Principles such as independent causal mechanisms and mechanism sparsity have proven essential as inductive biases, guiding models to distinguish between stable causal relations and transient statistical associations, thereby ensuring that learned representations are both identifiable and interpretable.

Methodologically, the landscape has transformed from discrete combinatorial search to continuous optimization and differentiable structure learning. This shift has enabled scalable causal discovery in high-dimensional regimes previously inaccessible to traditional algorithms. The emergence of specialized architectures, including Graph Neural Networks and diffusion-based generative models, has facilitated the handling of indefinite data types and complex relational dependencies. Furthermore, the integration of Large Language Models into causal pipelines marks a significant step toward hybrid systems that combine semantic reasoning with statistical rigor. These hybrid approaches address critical challenges related to cold starts, variable definition in unstructured domains, and the validation of discovered structures against expert knowledge, effectively bridging the gap between raw data and high-level causal concepts.

Despite substantial advances, significant challenges remain in standardizing benchmarks, handling indefinite data paradigms, and ensuring the robustness of causal representations in fully unsupervised settings. Tensions persist between the flexibility required for modeling complex real-world data and the rigid constraints necessary for theoretical identifiability. Future progress hinges on the development of unified frameworks that can seamlessly integrate amortized inference, dynamic structure learning, and human-in-the-loop validation. Ultimately, the convergence of causal theory and deep learning promises to deliver artificial intelligence systems that are not only predictive but also robust, fair, and capable of genuine reasoning about the generative processes that shape our world.